

The Role of Audio in the Diegesis of Virtual Reality Games

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ABSTRACT

While static media makes the distinction between diegetic and non-diegetic audio, the use of these concepts in interactive media, such as games, presents problems of classification. This is especially true for virtual reality where players embody virtual characters through their own bodily movements. This PhD research focuses on the diegetic aspect of audio in virtual reality and the effect that different approaches might have on players' sense of embodiment and willingness to suspend disbelief.

CCS CONCEPTS

• **Human-centered computing** -> **Human computer interaction (HCI)** -> **Interaction paradigms** -> **Virtual reality**; • **Human-centered computing** -> **Human computer interaction (HCI)** -> **Interaction techniques** -> **Auditory feedback**;

KEYWORDS

Virtual reality, audio, diegesis, embodiment, willing suspension of disbelief

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1 CONTEXT AND MOTIVATION FOR RESEARCH

Audio has long been a fruitful research avenue in HCI [20] with many studies focusing on its use from perspectives such as representing various types of information [14, 19]. Among various areas, virtual reality (VR) has generated interest in the use of audio, since VR affords many new avenues for the study of human-audio interaction, such as the fact that the player embodies a virtual body representation and thus perceives the virtual reality environment (VRE) as if from inside this environment.

Previous work on audio in VR has focused on the use of 3D/spatial audio, which can create localizable sound cues in virtual spaces [6, 23]. The unique qualities of audio in VR applications are also highlighted, including the omnidirectional and temporal

nature of sound perception, allowing the audio channel to constantly provide information about an environment, whether real or virtual [12, 23]. Previous work also highlights the unique role and challenges of music in VR games, including discussions regarding spatial audio and the relationship between music and presence [17, 18]. Static media, such as literature and film, distinguish between diegetic and extradiegetic/non-diegetic content, with diegetic content referring to that which originates from within a narrative world, such as dialogue, and non-diegetic content to that which originates from outside this narrative world, such as music [9]. However, despite ample amounts of studies focusing on different aspects of audio in VR environments, investigation regarding the “diegesis” of audio content in VR applications has not yet been thoroughly studied.

The topic of diegesis is of special importance in this context, since it is not only a contested issue for VR games [15] but also presents opportunities for entirely new uses of audio that are unique to the medium of VR [28]. Investigating the utilization of different ways of audio in VR environments (VREs) is particularly important because of its effect on kinesonic congruence, which can aid players in extending their body schema to embody virtual characters [3:57], its ability to help/harm the illusion of a coherent virtual world [16], and the effect on players' perception of the believability of VREs [18].

This PhD research will investigate the integration of audio in VR games by utilizing diegesis as a lens in the consideration of the roles of various audio types in VR games. In order to accomplish this, we will first review existing cross-disciplinary knowledge from previous audio-focused studies in VR to create implications for design. These implications will be used to create a prototype in the form of an experimental VRE that aims to explore new concepts relating to VR game audio. The developed prototype will be tested by users in empirical randomized and controlled experiments to understand their impact on users' experiences of embodiment, body ownership, and willing suspension of disbelief.

2 KEY RELATED WORK

2.1 Embodiment

One of the defining characteristics of VR is the notion of presence, i.e., the illusion of being physically transported to a VRE and thus of experiencing all events contained within this environment directly as if from a virtual body representation [25, 26]. A unique ability of VR systems in this regard is the ability to completely surround the player's field of regard, and much of their field of view, in terms of the various sensory stimuli they receive [2], such as allowing them to look in all directions and by providing spatial audio which utilizes psychoacoustic properties of 3-dimensional

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hearing to create audio in perceived 3D environments [29]. The fact that headsets track the position of players' heads also allows audio cues to be adapted as they move their heads, providing intuitive sensorimotor contingencies (SCs) for the perception of audio [25] and a stronger sense of "kinesonic congruity" between embodied actions and audio responses from the game system [3:35]. It should be noted that since these SCs require head-tracking, this project will focus on the use of head-mounted displays (HMDs) in VR.

This sense of congruity has major implications for embodiment in VR games, since emergent meanings between actions, audio, and visual cues can influence player behavior, especially with regards to gesture-controlled gameplay [3:37]. Furthermore, audio cues carry the connotation of haptic and visual sensations which, if executed with a sufficient degree of congruity between actions and audio cues, helps players extend their body schema, i.e., sensory sense of self, to include virtual characters as well [7:102]. This congruency between actions and audio cues has further implications for the design of audio cues, such as sound effects, music, and ambience, in the context of its diegesis [3:55,10], however, the extent to which aspects of diegesis might affect players' sense of embodiment is not clear [21, 27]. While the sense of embodiment is part of the experience of a wide variety of media experiences [27], it is a crucial component of VR experiences, since it provides a point of reference for users to experience the VRE as believable and place-like [25]. By studying the effects of audio diegesis on the sense of embodiment, this PhD project will produce knowledge that will add to the design space of audio design for VR game experiences.

2.2 Realism and disbelief

Although realism is a commonly cited design goal of video games and simulation based VRE [2], the concept of realism as verisimilitude to objective reality is often an insufficient way of defining what participants perceive to be "realistic" environments [1]. A more appropriate way to discuss this issue would be to use the concept of "willing suspension of disbelief", which relies on players' ability to overlook impossibilities and fictional elements of an experience [11]. In the context of VR, willing suspension of disbelief often manifests itself as part of the phenomenon known as presence [11], such as through the "place illusion" of being in a completely different space despite consciously knowing that you are not [25]. Presence measures are also used in VR research as part of the study of willing suspension of disbelief [13, 24].

In the context of willing suspension of disbelief, one could make the argument for the use of audio that mimics the real world in its representation, since players might expect audio to behave similar to real-world audio, including changing based on players' position and orientation [15]. The visual frame in VR also completely surrounds a player which allows both the audio and visuals to cohesively envelop a player. A notable counterargument against the use of diegetic audio relates to the fact that listeners tend to place much higher priority on timbral fidelity than spatial fidelity, i.e., on the quality and characteristic of the audio rather than its spatial reproduction, and it is argued that most listeners are not particularly concerned about spatial fidelity at all [22]. Since audio spatialization techniques incur a "cost" to timbral fidelity, it has been suggested that the result of slightly harming the perceived

fidelity of audio content could create a notable unease in the perception of said audio content, similar to the phenomenon known as the "uncanny valley" effect [22]. A proposed solution, similar to that of the more widespread uncanny valley effect, is thus to steer away from attempting perfect spatial fidelity to maintain timbral fidelity [22].

However, as mentioned, there is a lack of discussions that specifically focus on the role of suspension of disbelief in overcoming design issues of VR games, such as the diegetic functions of immersive audio. Games also present the potential for entirely unique diegetic functions of audio, such as player-generated audio, which also present interesting design challenges for the diegesis of VR games because of the unique role of audio in embodiment. Self-produced audio cues, such as "action sounds" produced by player movements [23], can create stronger connections between actions and audio outcomes, which may help players extend their body schema to virtual characters to "become" said characters [3:15]. However, this also creates the potential for kinesonic incongruence if self-produced audio cues, such as voice-overs, do not align with player actions [27]; the details and effects of different self-generated audio cues in games, however, is unclear [27]. By studying the effects of audio diegesis on willing suspension of disbelief, this PhD project will provide knowledge regarding the effect of audio design strategies on players' experience of VR games.

2.3 Diegesis

Although the distinction between diegetic and non-diegetic is often applied in the context of game audio, the interactive nature of games reduces the usefulness of this binary distinction, since many audio cues in games do not fall neatly into either category, thus making them "transdiegetic" [8] or "half-diegetic" [4]. The fact that VR games allow players to embody a virtual character and experience the environment as if from the embodied/egocentric perspective of this fictional character further complicates the distinction between the traditional notions of diegetic vs. non-diegetic, since players are fully surrounded by the sight and sound. Thus, players can be thought of as entering the diegesis of the narrative world themselves, by means of inhabiting a virtual body which surrounds their field of regard and from which they interact with the VRE, often with the use of gesture-based technologies. This can be contrasted with PC/console games where the player can always distinguish the physical boundaries of the game's displays and is therefore always "outside looking in" to a degree.

Thus, since the player is situated inside the diegesis of the game world along with all the visual content they perceive [15], there is some debate about which aspects of game audio should be diegetic or non-diegetic, e.g. such as whether game music or should rely on other strategies to incorporate musical content into the game world itself, such as using entirely diegetic music scores [17] or using musical strategies which blur the line between music and sound effects [5, 28]. Discussions regarding such fictional worlds/narratives are often tightly linked with the broader discussion regarding what is known as "willing suspension of disbelief", which is still an ongoing discussion in the area of traditional and VR game audio [18, 27]. By studying the effects of diegesis on embodiment and willing suspension of disbelief of VR games, this PhD project will contribute

to knowledge regarding the use of diegesis in sound design for VR games.

3 SPECIFIC RESEARCH OBJECTIVES, GOALS OR QUESTIONS

Main research question: how can the role of audio in VR games be positioned to reflect the game's diegesis?

Sub research questions:

- What is the state of the art in the literature investigating immersive audio in VREs?
- How can sound design in VR games be approached in terms of diegetic and non-diegetic audio?
- What is the effect of different diegetic strategies for VR game audio on players' perception of embodiment of their avatars?
- What is the effect of different diegetic strategies for VR game audio on players' willing suspension of disbelief?

3.1 Research progress to date

Data analysis on the corpus of research regarding the effect of audio on user experience has already been completed and a paper based on this analysis has been submitted for review.

3.2 Research approach, methods and rationale

The project takes the form of an article-based PhD with four planned publications to answer the research questions.

- The first publication takes the form of a systematic literature review to establish the state of the art in audio for affecting the user/player experience (UX/PX) in VR. By investigating how audio has been found to influence various aspects of UX/PX, this was expected to provide a theoretical framework for proceeding with the rest of the project as well as provide insight and an agenda for future research to be done in this direction.
- The second publication makes use of a research-through-design approach to iteratively create prototypes that explore possible roles of audio and diegesis in VR games. Building on the theoretical framework established through the first publication, the primary outcome of this publication will take the form of an interactive VR prototype that demonstrates strategies for using audio with regards to its diegesis.
- Using the prototype created for the second publication and outcomes to be studied from the first publication, the third publication makes use of a laboratory experiment to investigate the effect of diegetic audio strategies on psychological outcomes related to users' sense of embodiment. By exploring different audio categories and different diegetic strategies, the overall outcomes will be studied in a 2x2 factorial design.
- Similar to the previous publication, the fourth publication aims to use the prototype to investigate the effect of diegetic audio strategies on psychological outcomes related to users' willing suspension of disbelief, possibly through measures such as presence. These will also be studied in a 2x2 factorial design.

As such, the project aims to delineate the problem space of audio diegesis in VR by establishing the current state of the research

and using this to design possible solutions. The effect of these solutions will then be tested experimentally with a robust research design to provide concrete insight into how audio might be designed for VR with regards to its diegesis and how this might affect the experiential aspects of users/players.

4 RESULTS AND CONTRIBUTIONS TO DATE

Results from the first publication (which is still under review) have provided insight into the current state of research in audio for VR and highlights the ability of audio to influence a variety of experiential outcomes, such as pleasantness and realism, but also points out outcomes where this is less clear, such as influencing presence. Factors that might influence results were also pointed out, such as novelty bias, research design, a lack of standardization of concepts and research instruments, and technological differences.

5 EXPECTED NEXT STEPS

Progress is being made on another paper based on the remaining data from the data analysis of the corpus toward another paper. This includes qualitative experience data specifically related to audio diegesis. After that is submitted, work is planned to begin for the second publication by using insights gleaned from the previous publication to design and implement the VR prototype described above.

6 DISSERTATION STATUS AND LONG TERM GOALS

My long-term goal is to pursue a career as a researcher and lecturer in South Africa. Through completion of this degree, I hope to gain a solid understanding of the research process as well as the domain areas I am doing research in. With this knowledge I would be able to help build research capacity in South Africa for niche subject areas, such as VR, game design, sound design, and more broadly for HCI. This would be beneficial, since supervisory capacity for domain areas such as these is generally lacking in South African universities even though there has been much interest in using VR in various lucrative domains, such as mining engineering.

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